

TASK 2 — Virtual Try-On AI Models

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1. Aim of the Study

The purpose of this task was to study how Artificial Intelligence is used in modern virtual fitting and digital fashion systems. The research focused on understanding how AI can detect human body posture, analyse body shape, and display digital clothing on users in real time.

This task also explored how virtual fitting technology improves online shopping platforms and fashion recommendation systems.

2. Introduction to AI Virtual Fitting Systems

AI virtual fitting systems are intelligent computer vision applications that allow users to digitally try clothes before purchasing them online. These systems use Artificial Intelligence, Machine Learning, Pose Detection, and Augmented Reality to create interactive shopping experiences.

Modern fashion platforms use virtual fitting technology to:

- display clothes digitally
- improve customer experience
- reduce clothing return rates
- increase shopping confidence
- provide realistic clothing visualization

Virtual fitting systems are becoming highly important in:

- fashion e-commerce
 - online clothing stores
 - digital fashion platforms
 - virtual shopping applications
 - AI styling systems
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3. Technologies Used in Virtual Fitting Systems

The following technologies are commonly used in AI virtual fitting applications:

Technology	Purpose
Artificial Intelligence	Smart body analysis
Computer Vision	Human body detection
Pose Estimation	Body landmark tracking
Body Segmentation	Separating body from background
Augmented Reality	Digital clothing overlay
Deep Learning	Real-time prediction models

4. Working Process of AI Virtual Fitting Systems

Step 1 — Human Body Detection

The system first detects the person using AI computer vision algorithms.

The AI model identifies:

- face
- shoulders
- arms
- torso
- legs

This helps the system understand the body structure.

Step 2 — Pose Landmark Tracking

Pose detection models identify body landmarks such as:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

These landmarks are used to estimate body posture and movement.

Step 3 — Body Segmentation

The AI system separates the user from the background using segmentation algorithms.

This improves:

- clothing alignment
 - body shape estimation
 - fitting accuracy
 - real-time tracking
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Step 4 — Clothing Overlay

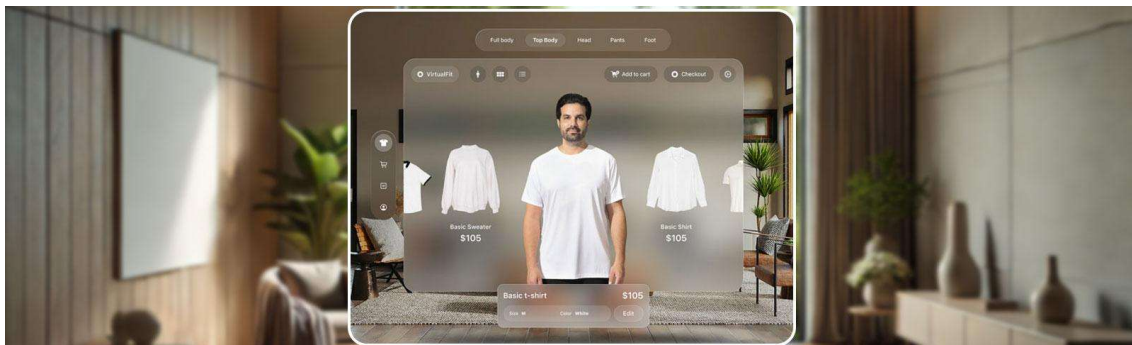
Virtual clothing is placed over the body using AI alignment techniques.

The system adjusts:

- clothing size
 - body movement
 - body posture
 - cloth positioning
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Step 5 — Real-Time Visualization

The user can move in front of the camera and observe live virtual clothing fitting in real time.



5. Applications of AI Virtual Fitting Systems

AI virtual fitting technology is widely used in:

- online clothing platforms
 - virtual shopping systems
 - digital fitting rooms
 - eyewear try-on systems
 - AI fashion assistants
 - augmented reality applications
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6. Advantages of AI Virtual Fitting Systems

Virtual fitting systems provide multiple advantages:

- improved shopping experience
- better clothing visualization
- reduced product returns
- accurate clothing size estimation
- increased customer confidence
- personalized shopping experience
- faster product selection



7. Challenges in Virtual Fitting Technology

Challenge	Description
Lighting Problems	Poor lighting affects body detection
Loose Clothing	Large clothes hide body shape
Background Complexity	Complex backgrounds reduce accuracy
Real-Time Processing	High computation may reduce performance
Body Occlusion	Hidden body parts reduce tracking quality

8. AI Models Used in Virtual Fitting

Popular AI frameworks used include:

- MediaPipe
- OpenPose
- TensorFlow
- OpenCV
- PoseNet
- DeepLab

These frameworks help in:

- pose estimation
 - body tracking
 - segmentation
 - AI clothing alignment
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9. Learning Outcomes

During this task, the following concepts were learned:

- AI virtual fitting workflows
- pose estimation systems
- body segmentation methods
- clothing overlay techniques

- AI fashion applications
 - computer vision systems
 - real-time body tracking
 - augmented reality workflows
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10. Conclusion

AI virtual fitting systems are transforming the fashion industry using Artificial Intelligence and computer vision technologies. These systems improve online shopping experiences by allowing users to digitally try clothes before purchasing them.

The research improved understanding of:

- virtual fitting systems
- AI fashion technology
- pose estimation
- body segmentation
- computer vision workflows
- intelligent shopping systems

